

PATRICK LESSA

Senior Data Engineer

Python · SQL · Apache Airflow · Spark/PySpark · Cloud Data Pipelines · ETL/ELT

Florianópolis, Brazil (Remote) · patrickwbarbosa@gmail.com · English: Advanced (B2–C1)

PROFESSIONAL SUMMARY

Senior data engineer with 8+ years designing, building, and owning production-grade data pipelines, ETL/ELT workflows, and analytical platforms that power executive and operational decisions. Strong hands-on background in Python, SQL, Apache Airflow, Docker, and Spark/PySpark across cloud data environments (Microsoft Fabric / Lakehouse, Azure SQL). Proven record orchestrating daily, multi-country datasets with reliability, anomaly detection, and thorough documentation. Applied GenAI / RAG background (embeddings, vector search, observability) for AI-native data products. Available for remote, contract engagements.

CORE SKILLS

Data Engineering: ETL/ELT, data pipelines, multi-source ingestion, workflow automation, data validation, migration, orchestration

Languages: Python (Pandas, NumPy, PySpark), advanced SQL

Orchestration & Infrastructure: Apache Airflow, Docker, Spark / PySpark, Git

Cloud & Platforms: Microsoft Fabric (Lakehouse, Dataflows Gen2, Spark notebooks), Azure SQL; AWS (foundational, expanding); GCP / BigQuery–aligned

Data Stores: SQL / Azure SQL, PostgreSQL / Supabase, pgvector (vector search)

Data Modeling: dimensional modeling, star schema, KPI / metric layers, semantic models, performance tuning

APIs & Integration: REST APIs, SharePoint, enterprise systems (Zeus ETRM)

Applied AI / Data: RAG, embeddings, LangChain, semantic & hybrid retrieval, LLM workflow observability

Delivery: requirements translation, technical documentation, stakeholder communication, remote / distributed collaboration

PROFESSIONAL EXPERIENCE

Senior Data & Analytics Engineer (official title: Power Platform Developer)

Jul 2024 – Present

Mouts TI · Remote

Clients: Atlas Renewable Energy (LATAM energy trading), ENGIE, Starrett, credi&gente

- Build and own production-grade data pipelines with Apache Airflow and Docker, orchestrating daily ingestion, transformation, and delivery of multi-country market and valuation datasets, with automated anomaly checks in the first 30 minutes of each run.
- Automated ingestion of 14 SharePoint and SQL sources, cutting monthly data preparation from 6 hours to 45 minutes (~85% efficiency gain).
- Engineered Python integrations against REST APIs, SharePoint, and the Zeus ETRM platform, replacing manual routines with reliable, observable, and documented workflows.
- Developed Spark / PySpark and Lakehouse transformation logic in Microsoft Fabric (Dataflows Gen2, Lakehouse, Spark notebooks) for forecasts, price curves, and curtailment datasets, improving standardization and analytical readiness.
- Modeled scalable analytical layers with star-schema design and country-level row-level security, improving the reliability and scalability of reporting for LATAM operations.
- Migrated backend data structures from SharePoint Lists to Azure SQL and adapted downstream applications, improving reliability, scalability, and performance.
- Authored technical documentation for data models, transformations, metrics, and business rules to support governance and onboarding across distributed teams.

Data Analyst & Project Manager

Apr 2023 – Jul 2024

TD Business · Remote

- Co-led delivery of 100+ BI dashboards and analytical solutions for public agencies (MEC, Brazilian Space Agency, FEESC, Sebrae), owning measurement systems end-to-end — from requirement framing to KPI definition to delivery.
- Translated loosely defined business needs into indicators, datasets, and reporting logic that became the canonical source of truth for stakeholders.
- Structured the Sebrae Startups Observatory analyses for innovation-ecosystem mapping and recurring executive reporting.

Junior Data Analyst

Feb 2023 – Apr 2023

Stemmer Foundation (FEESC) · Remote

- Mapped and instrumented 363 institutional processes; designed 20+ strategic KPIs and automated 10 manual spreadsheets, compressing update cycles from weeks to days.

SELECTED PROJECT

Xplore Dados / Smartchat — Founder & Developer

Ongoing

Independent AI & Data Product

- Built Python data pipelines for a RAG-based research product: multi-source ingestion, document processing (PDF/DOCX), vector search with pgvector, caching, and fallback logic.
- Applied observability (latency, cost, answer quality) and multi-provider LLM orchestration (OpenAI, Gemini, Claude); backend on Supabase / PostgreSQL.

EDUCATION

Ph.D. in Business Administration — UFSC (in progress)

M.Sc. in Business Administration — UFSC (CAPES scholarship)

B.A. in Business Administration — UFCA (PIBIC/CNPq scholarship)

Focus across degrees: quantitative methods, segmentation, and Python-based research.

CERTIFICATIONS & LANGUAGES

Certifications: Google Data Analytics · Python for Data Analytics · Supervised Machine Learning · Power BI · Advanced Excel

Languages: Portuguese (native) · English (advanced, B2–C1) · Spanish (basic) · French (basic)